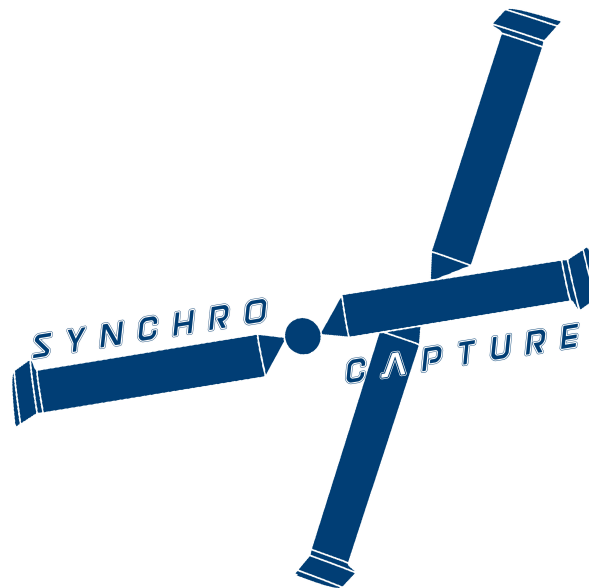


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

Weight Sizing and Subsystem Design of a Novel Synchropter Concept for the Aerial-Recovery of Falling Rocket Stages



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Abstract

Mid-air recovery of falling rocket stages offers a substantial cost reduction potential for orbital missions. An unmanned, remote piloted helicopter was designed to retrieve multiple stages of a rocket. Preliminary weight sizing involved a linear regression analysis based on existing unmanned rotorcraft. Individual component weights were evaluated based on AFDD models. Obtained **empty weight is 1228.5kg**, which with the 1400kg payload and 278.1kg of fuel gives **the maximum takeoff weight of 2906.6kg**. An S-duct inlet with **0.125m² intake area** and a straight **exhaust duct with 0.0375m² cross-section area** were designed for the turboshaft engine with a **dry-sump oil system**. A system consisting of four **hydraulic arms and a 16.28m × 4.5m nylon cargo net** is designed to capture the payload during landing.

WRR Team Members

Table 1: *Team Members and Roles.*

Name	Team Name	Team Code
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Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
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†Sub-team leads are shown in **bold**

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Part I

Conceptual Design

1 Introduction

Mid-air recovery concept has been developed for decades [1], but the demand for it grows nowadays with an increasing number of planned orbital missions [2]. Mid-air retrieval creates an opportunity for long term cost-reduction and more sustainable space launches. Consequently, a need exists for a new vehicle capable of recovering multiple stages of a rocket. The mission involves a cruise to the recovery site and a capture manoeuvre. An unmanned helicopter was selected for the mission at the conceptual design stage [3]. The initial weight sizing of the recovery vehicle is summarised in this report, together with possible inlet and exhaust arrangement for the turboshaft engine, the oil system and a capture system for landing on the boat.

Part II

Preliminary Design

2 Weight Sizing

The objective of the weight sizing process was to find the maximum takeoff weight (MTOW) of a novel rotorcraft design for the recovery of a falling rocket stage. The aim was to ensure that the design is as light and efficient as possible, while being able to meet all mission requirements, in particular the payload capacity of $W_{pl} = 1400\text{kg}$ and endurance $t = 2\text{h}$. These parameters were used to initialise the weight sizing process.

This report puts emphasis on certain aspects of the sizing process, in particular the historical data analysis and individual component weights estimation. Remaining information can be found in another report on weight sizing [4], exploring the Tishchenko methodology and power and fuel weight estimation in more detail.

2.1 Historical Data Analysis

Initial approach to the weight sizing process was to perform the linear regression analysis based on available data on unmanned helicopters. Trends in the historical data were used to predict parameters of a new helicopter capable of carrying an external payload of 1400 kg.

2.1.1 Existing Unmanned Rotorcraft

After an initial review of various existing unmanned helicopter designs, it was decided that regression analysis should be narrowed down to unmanned rotorcraft designed to carry an external slung load. This was to ensure that analysed vehicles match mission requirements and all estimates are as accurate as possible. Data gathered for the regression analysis [5] [6] [7] is summarised in Table 5 in Appendix A.1.

First estimates were based solely on the vehicle payload carrying capacity, as shown in Figure 1.

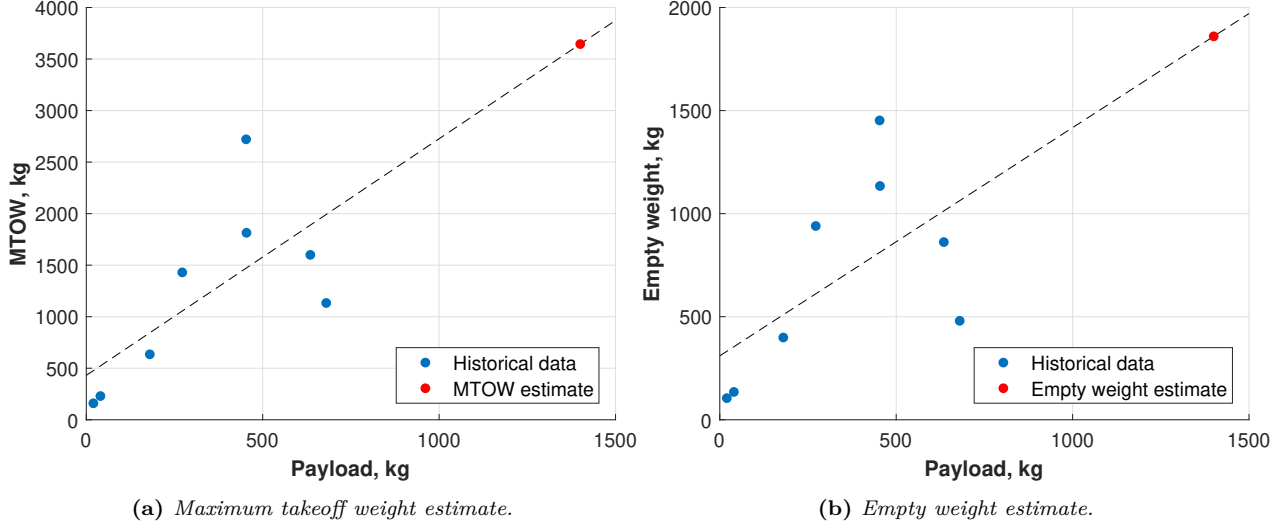


Figure 1: First weight estimates for a payload of 1400 kg, based on existing unmanned helicopters.

Obtained values for the maximum takeoff weight and empty weight were $W_{to} = 3645\text{kg}$ and $W_{empty} = 1860\text{kg}$, respectively. These results are very approximate, since they are based on a small dataset of only 8 existing helicopters with a payload capacity significantly lower than 1400kg. While this MTOW estimate is very rough and cannot be considered reliable on its own, it was used as an input to the initial gross weight estimation procedure given by Prouty [8].

2.1.2 Prouty Sizing Methodology

Following steps were taken to obtain an improved MTOW estimate, as suggested by Prouty:

1. Initial estimates for the MTOW and installed engine power P_i are made based on historical data. Figure 2 illustrates the installed power regression analysis, for $W_{to} = 3645\text{kg}$ it yields $P_i = 734\text{kW}$.

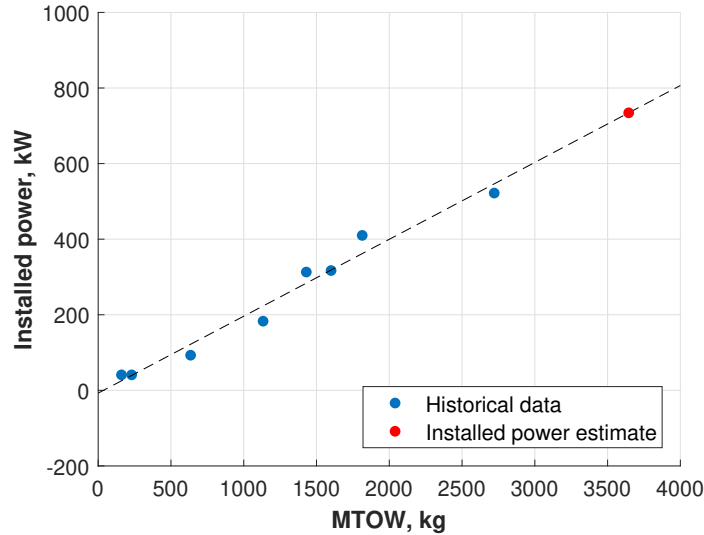


Figure 2: Installed power estimate for a MTOW of 3645 kg, based on existing unmanned helicopters.

2. Required fuel weight is obtained from

$$W_{fuel} = sfc \times P_i \times t, \quad (1)$$

where $sfc = 0.4\text{lb/h.p.hr}$ is the specific fuel consumption of a gas turbine.

3. Useful load W_{ul} is calculated as the sum of W_{pl} and W_{fuel} .
4. Ratio W_{ul}/W_{to} is estimated from existing unmanned rotorcraft as 0.5.
5. New MTOW estimate is calculated as

$$W_{to} = \frac{W_{ul}}{W_{ul}/W_{to}} \quad (2)$$

and the process repeats until convergence is reached.

It should be noted that the convergence criterion was set as 1%. The iterative process gave a new estimate $W_{to} = 3495\text{kg}$, therefore 4.2% lower than the initial approximation which did not involve the power and fuel weight estimation.

2.2 Empty Weight Estimate

While the regression analysis was a good starting point, a need for a more precise method arose when the baseline configuration was confirmed and following points had to be considered [9] [3] [10]:

- Since the helicopter is unmanned, the cockpit is redundant. As a result, the fuselage can be reduced to about 40% of its typical size.
- Subsystems such as: cockpit controls, auxiliary power unit, instruments, furnishings and environmental controls were eliminated from the design [11]. Remaining subsystem weights were treated as constant in the sizing process.
- The design does not include a tail rotor.

To account for these unconventional design choices, having consulted multiple University of Maryland reports on novel helicopter design [12] [13] [14], it was decided that the weight sizing process should involve an empty weight estimation done by summing individual weights of all components present in this particular helicopter. Aeroflightdynamics Directorate methodology (AFDD82) was used to obtain component weights. Estimations were made using parametric equations derived from existing rotorcraft powered by gas turbines, listed in Chapter 19 in [15]. Baseline configuration dictated which parts should be accounted for in the total empty weight.

2.3 Iterative Weight Sizing Process

Apart from the mission profile [9] and payload weight, the sizing algorithm required a number of fixed inputs. These were provided by the Aerodynamics team [16] and are given in Table 2.

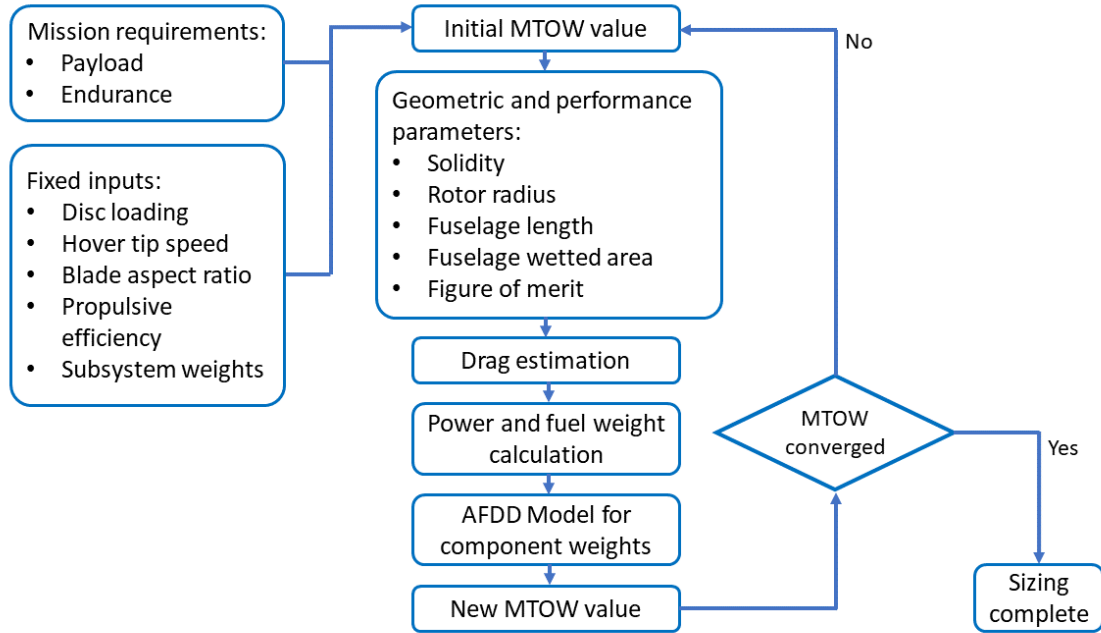
Table 2: *Fixed inputs for the initial weight sizing.*

Rotor disc loading, kg/m ²	Rotor hover tip speed, m/s	Blade aspect ratio
6	221	18

The above values were used to initialise the algorithm, which consisted of following steps:

1. A starting estimate for the MTOW value is made. For the first iteration, a value of 3600kg was taken.
2. Parameters of the main rotor and tail surfaces are calculated [16] [17] [18].
3. Vehicle drag estimation is updated [19].
4. Power required [20] and fuel weight [4] per each segment are evaluated.
5. Weight of each component is updated for new geometry and performance parameters. Empty mass is calculated.
6. Payload, fuel and empty weight are added to obtain a new MTOW value. The process repeats if not converged.

The algorithm is illustrated in Figure 3.

**Figure 3:** *Summary of the iterative initial weight sizing process.*

The process was repeated until the difference between initial and final MTOW values was less than 0.5%. It is important to note that during the weight sizing process, the payload weight was included in maximum takeoff weight, even though in practice it would be captured in flight. This assumption was made to ensure that the most extreme case of taking off with maximum fuel and payload is considered.

2.4 Results

Results of the weight sizing process are summarised in Table 3 below.

Table 3: *Component weight estimates.*

Component	Weight, kg	Weight, lbs	% Empty Weight
Rotor Group	167.2	368.3	13.6%
Blades	99.7	219.6	
Hub	49.5	109.0	
Spinner	18.0	39.7	
Empennage Group	18.8	41.5	1.5%
Horizontal tail	14.9	32.8	
Vertical tail	3.9	8.7	
Fuselage Group	68.8	151.5	5.6%
Basic structure	64.9	142.9	
Crashworthiness	3.9	8.6	
Lighting Gear Group	116.0	255.4	9.4%
Basic landing gear	94.2	207.4	
Retraction	7.5	16.6	
Crashworthiness	31.4	14.3	
Propulsion Group	267.6	589.5	21.8%
Engine	207.3	456.6	
Accessories	34.2	75.4	
Nacelle	26.1	57.5	
Transmission Group	221.7	488.3	18.0%
Gear box	188.1	414.3	
Rotor shaft	28.1	62.0	
Drive shaft	0.7	1.6	
Rotor brake	4.7	10.4	
Subsystems	368.4	811.4	30.1%
Flight controls group	42.8	94.2	3.5%
Electro-hydraulic group	79.5	175	6.5%
Electrical group	45.4	100	3.7%
Avionics group	30.0	66.0	2.4%
Anti-icing group	38.6	85.0	3.1%
Load and handling group	68.1	150.0	5.5%
Fluids	22.7	50.0	1.8%
Fuel system	41.4	91.2	3.4%
Empty weight	1228.5	2705.9	100.0%
Useful load	1678.1	3699.5	
Fuel	278.1	613.1	
Payload	1400	3086.4	
Maximum Takeoff Weight	2906.6	6407.8	

2.5 Discussion

To verify the accuracy of the results, the share of each component in total empty weight was compared to a representative empty weight breakdown of a helicopter, as seen in Figure 4 [21].

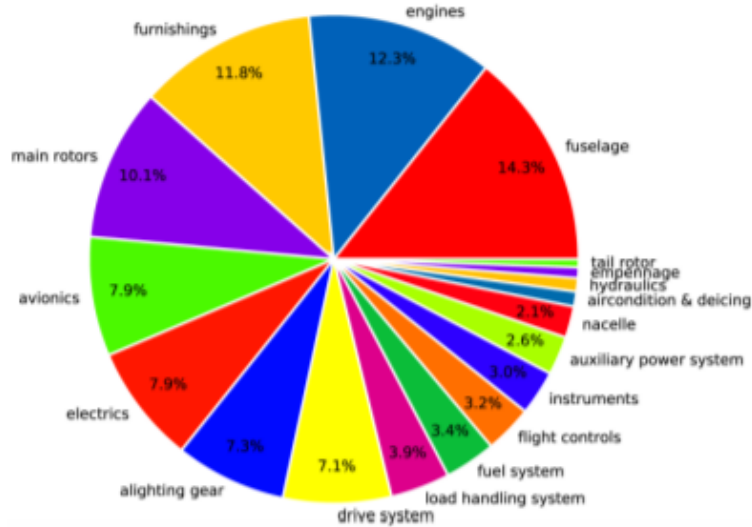


Figure 4: Typical rotorcraft empty weight breakdown by component, based on AFDD models.

An obvious discrepancy between the two breakdowns is the difference in the percentage share of the fuselage weight. Usually the heaviest of all component groups (14.3%), in this case is reduced to 5.6%. This is in accordance with what was expected, considering that majority of the fuselage was removed from the design since the vehicle is unmanned. This is also reflected in the final MTOW value of 2906.6kg, which is 17% lower than the previous estimate based on Prouty.

Propulsion group and drive system percentage shares, however, are higher than typical - 21.8% vs 12.3% and 18.0% vs 7.3%, respectively. This stems from the fact that while other components could be reduced in size, the powerplant had to be sufficiently large for the vehicle to satisfy mission requirements and payload capacity. Remaining component weights calculated from parametric equations are similar to representative values in Figure 4.

It should be highlighted that although the AFDD methodology for empty weight estimation greatly improved the accuracy of the result by allowing to only include parts that are present in the chosen configuration, it nevertheless relies on historical data and consequently will never be fully adequate for a novel, unconventional design. For a dataset consisting of 42 helicopters, average error in total weight is 5.3%. For the structural components alone the error is 6.1%, while for the propulsion group it is 10.9% [15]. This again justifies the unusually high percentage share of the powerplant.

Obtained weights were used as a starting point for the detailed design of helicopter components. Foreseeably, some of the parameters used as inputs to the AFDD model were updated as the design was known in more detail. Furthermore, as the vehicle configuration was updated in certain aspects, some components previously included in the calculations became irrelevant - in particular, the decision to move the engine inside the fuselage allowed for a reduction in weight by removing the nacelle. Updated component weights are included in a separate report on weight sizing [4].

3 Engine Components

Contrary to what is seen in conventional rotorcraft designs, the engine was moved from its standard location to the inside of the fuselage. This is due to the fact that the cockpit is removed from the design, hence the space in the fuselage is available for the powerplant. Furthermore, the engine is inclined in order to shift the centre of gravity forward. These choices largely affect possible inlet and exhaust configurations.

3.1 Inlet

Finding an appropriate inlet arrangement posed a significant challenge as it had to fit around other components in a relatively small fuselage. Following constraints on the inlet placement had to be considered:

- The position of the gearbox, placed directly under the two rotors, cannot be changed [22].
- The inlet cannot be located close to the ground to prevent engine contamination or foreign object damage.
- The intake should have such shape that it does not add significant drag to the fuselage.
- The duct should be straight if possible, or curved gradually to minimise losses caused by flow separation and inlet pressure distortion [23].

To satisfy the above requirements, three options were considered:

- Straight inlet duct: First proposed arrangement was a nearly vertical straight duct with an intake between the two rotors. It was rejected, as in order to allow the air to enter in forward flight, the duct would have to turn abruptly near the top of the fuselage. This would cause pressure losses and outweigh the advantage of having a straight duct.
- Side intake: Another option considered was to make two flush NACA inlets on the side of the fuselage and as a result have two ducts that pass the gearbox on its sides and then are joined right before they reach the engine. This would be beneficial aerodynamically, as the fuselage shape would remain unaffected. However, placing the inlet on the side, and as a result closer to the ground, increases the risk of dust or sea water contamination. There is also no benefit of the air being pushed into the engine in forward flight. Furthermore, splitting the duct in two adds complexity to the shape and causes greater losses. This idea was therefore discarded.
- Top of the fuselage: A final decision was made to place the intake between the rotors, then connect it to the engine by a gradually curved, single S-duct passing above the gearbox. It was accepted as it reduces the risk of foreign object damage and allows the air to be pushed in during forward flight.

3.1.1 Inlet Geometry

Following the guidelines from the Aerodynamics team [19], it was decided that the inlet opening should be elliptical to minimise drag. From the propulsion efficiency perspective, the optimum Mach number of the flow as it enters the compressor with area $A_c = 0.043\text{m}^2$ is $M_c = 0.55$ [20]. The duct therefore converges to accelerate the flow from freestream conditions, i.e. $M = 0.16$ in cruise. Using 1D Compressible Flow Relations [24], and accounting for approximately 2% losses in the duct [25], obtained inlet area is $A_i = 0.125\text{m}^2$ and the ellipse dimensions are $0.08\text{m} \times 0.50\text{m}$.

To maximise the pressure recovery in the S-duct, the first bend, which contributes the most to the pressure loss, should have a maximum possible radius of curvature [25]. This was applied in the design and the final configuration is illustrated in the CAD cross-section view in Figure 5 [26].

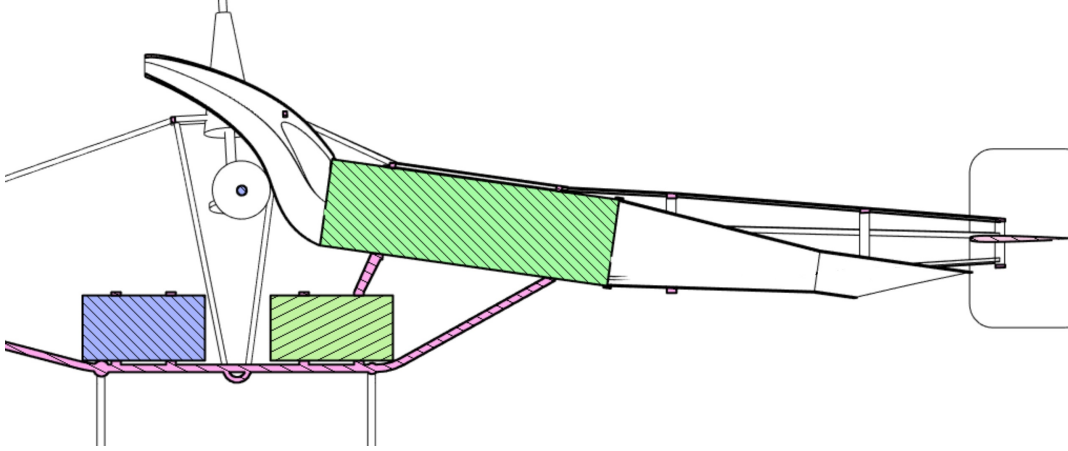


Figure 5: *Inlet and exhaust final configuration.*

3.2 Exhaust

Exhaust arrangement is displayed in Figure 5. Since the engine is mainly not producing thrust but shaft power, the primary concern in the exhaust design is not the speed but the temperature of the flow [27], which reaches 1000K. Therefore main guidelines to the exhaust placement are [28]:

- The exhaust gas must be directed away from aerodynamic surfaces or the capture mechanism to avoid damage.
- The exhaust should be placed outside the rotor wake to avoid exhaust gas reingestion.

With these constraints in mind, the exhaust was placed at the bottom of the fuselage, which was a natural consequence of having an inclined engine. The hot gas will not interact with the recovery mechanism or critical aerodynamic surfaces. Spacing between the vertical surfaces of the H-tail is sufficient to ensure they are not exposed to high temperature. A straight duct will be used for pressure recovery and ease of manufacture.

3.2.1 Exhaust Geometry

The gas pressure at the turbine exit, where the flow Mach number is approximately $M_t = 0.5$, is 4% higher than ambient [20]. The duct is therefore converging for a slight drop in pressure to match that of the freestream. Considering again 1D flow properties [24] and assuming minor losses in a straight exhaust duct with smooth walls [27], exit Mach number $M_e = 0.53$ was obtained and for the turbine exit area $A_t = 0.0395\text{m}^2$ the optimal exhaust area is $A_e = 0.0375\text{m}^2$. The duct has a circular cross-section with diameter $d_e = 0.219\text{m}$.

Details of the inlet and exhaust geometry calculations can be found in Appendix A.2.

Exhaust thrust T necessary for drag estimation was approximated as [29]:

$$T = \dot{m}v_e, \quad (3)$$

where $\dot{m} = 3.69\text{kg/s}$ is the engine mass flow rate and $v_e = 181.0\text{m/s}$ is the exhaust velocity. Obtained value of $T = 667.9\text{N}$ is 2.35% of the main rotor lift in hover, which is assumed to be equal to maximum weight i.e. 28514N. This ratio falls within the 2-4% range, typical for existing helicopters [30].

3.2.2 Material Selection

Exhaust material selection was dictated by temperature considerations. Selected material must sustain at least 1000K (727°C). To choose an appropriate material, a relationship between high service temperature and density was first considered. The search was further narrowed down by specifying a minimum corrosion resistance, namely by setting the minimum pitting resistance equivalent number (PREN) to 40, which is essential for a component exposed to hot exhaust gases [31]. Alloys considered at this stage can be found in the Appendix A.3. Selected HAYNES HR-120 Nickel Super Alloy, properties of which are summarised in Table 4 (based on the Cambridge Engineering Selector), has the lowest density of all acceptable nickel alloys and is more corrosion resistant than stainless steels.

Table 4: *Properties of the nickel alloy selected for the exhaust duct.*

Tradename	Density, kg/m^3	Maximum Service PREN Temperature, °C	Price, GBP/kg
HAYNES HR-120	$7.87 - 8.27 \times 10^3$	647 – 1040	25.4 – 44.2 7.27 – 9.27

3.3 Oil System

The oil system of a turboshaft engine serves two main purposes [32]:

- Cooling - The oil supplied to and scavenged from the engine enables a heat exchange, since a layer of oil acts as a heat transfer path. It absorbs excessive heat from engine shaft bearings and thus protects them from damage caused by high temperature. Hot engine oil is then cooled and filtered before reentering the bearing sumps.
- Lubrication - Oil extends the lifetime of engine components by protecting them from wear and friction, as well as corrosion and rust, which is particularly important when the helicopter is not used regularly. Lubricants are also used to ensure that all dirt is removed from engine parts and that cavities are sealed properly.

The system is therefore essential for protecting the engine from a catastrophic failure. There are two types of engine oil systems: wet and dry sump. In a wet sump system, the engine contains a built-in oil tank, while in the dry sump system it is stored in an external reservoir. For this helicopter design, a dry sump oil system is selected, as it gives an increase in power due to a reduction in viscous drag and pressure losses [33]. Since the oil coming from the external tank passes through a pressure regulating valve and is supplied to the engine at a constant pressure, the reliability of the powerplant performance increases.

The dry sump oil system contains following components, laid out as shown in Figure 6a below:

1. Bearing sumps - this is where the oil is supplied to and removed from when the engine is running, four compartments (shown in Figure 6b) are distributed along the shaft.
2. Scavenge filter - separates dirt and debris from scavenge oil.
3. Scavenge oil pump - removes hot scavenge oil from the bearing sump.
4. Air/oil separator - separates air bubbles carried by the oil after it is churned in the filter before it returns to the reservoir.
5. Oil reservoir - a tank for storing hot oil, attached to the engine casing. It is pressurised to maintain the flow to the pressure pump, and has a 10% extra space for oil volume expansion in increased temperature, as specified by Federal Aviation Regulations [34].
6. Pressure pump - pressurises the oil, powered by the engine.
7. Filter - filters particles from the pressure oil. Oil can by-pass the filter in case it is clogged.
8. Cooler - adjusts the oil temperature to appropriate levels by transferring the heat to the fuel.
9. Pressure regulator - adjusts oil pressure before it reenters the engine and is sprayed onto bearings.

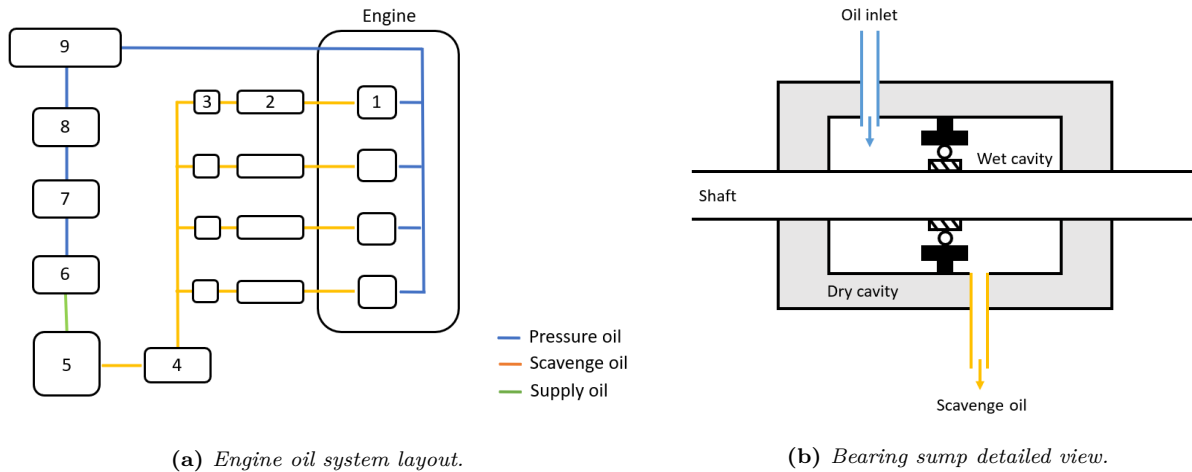


Figure 6: First weight estimates for a payload of 1400 kg, based on existing unmanned helicopters.

The AeroShell W 15W-50 ashless-dispersant oil is selected, as it is a semi-synthetic, oxidation-resistant oil suitable for very high temperatures observed in helicopter engines [35]. The high viscosity index ensures that the oil has enough load carrying capacity when it is hot [36].

Gas turbine shafts are normally supported by at least two bearings [37]. The turboshaft designed for this mission has two shafts [20], hence there are four bearing oil compartments. Presented detailed view of the bearing sump is based on a layout proposed by ExxonMobil [38]. To keep the oil in the wet cavity of the bearing sump and prevent leaking, it is surrounded by a pressurised dry cavity.

Regulations specify that for every 40 US gallons of fuel there should be 1 gallon of engine oil provided [39]. As fuel volume required for this mission is 77.81gal, the appropriate oil volume is $V_{oil} = 1.95\text{gal}$ (0.007 l) and for the density $\rho_{oil} = 800\text{kg/m}^3$ it gives mass $m_{oil} = 5.83\text{kg}$. It was assumed that the total system weight should be approximately twice the weight of the oil itself, based on another helicopter design with a similar engine weight [12]. Therefore the oil system weight is estimated as $W_{oil} = 12\text{kg}$. This is approximately 4.48% of the estimated propulsion group weight, which is similar to existing designs with a turboshaft [12], [40].

4 Landing Capture Mechanism

In order to successfully complete the mission, a capture mechanism is designed for the boat, where the rocket stage can be safely disposed of. This is done with the use of hydraulic arms. Initial idea to hold the rocket directly in the claws of two robotic arms was discarded due to high risk of squashing the rocket. Instead, a system with a cargo net onto which the payload can be released was designed. Wrapped in the net, the rocket is prepared for further transport.

4.1 Cargo Net

The system is sized for the most limiting case, i.e. the first stage of the rocket, with diameter $d = 1.50\text{m}$ and length $l = 14.28\text{m}$, given by the Launch Team. The net is then sized with a 1m margin on each side longitudinally, giving total length of 16.28m, while the calculated width is 4.5m to protect the payload from sliding to the side. At each corner of the net, there is a 7.20m long lanyard. One of these ends with a master link, while remaining three have self-closing hooks so that four cables can be connected. Existing helicopter cargo nets were researched. The load carrying capacity of medium-duty nets matches the 1400kg requirement of this design. Based on their specifications [41], a nylon net is selected.

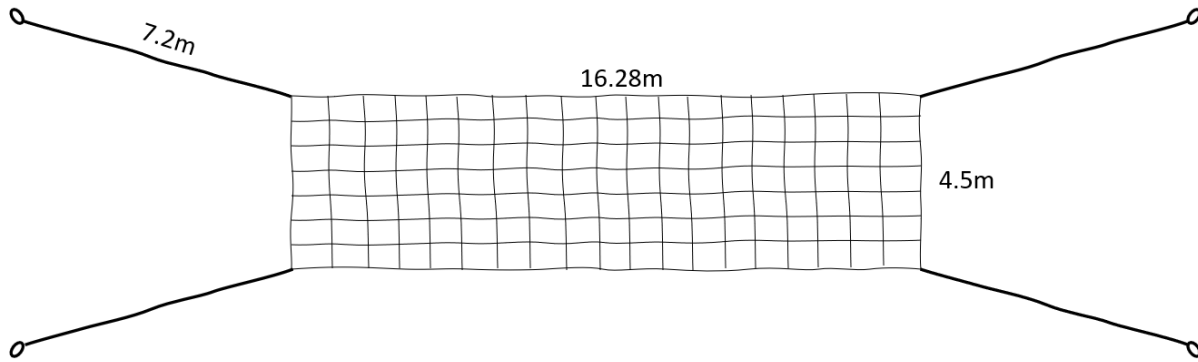
4.2 Hydraulic Arms

The hydraulic arms holding the net are sized and spaced based on two constraining cases:

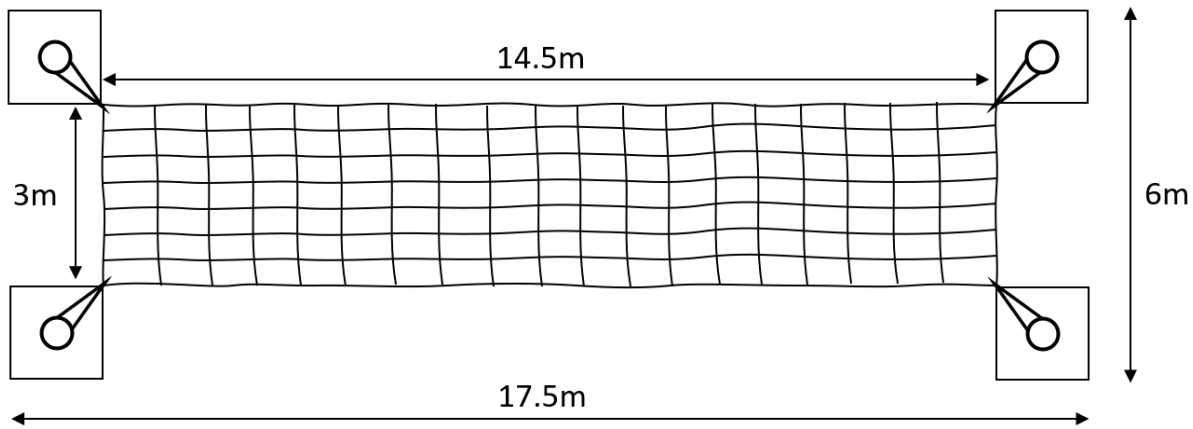
1. Before the landing, the cargo net is hanging from folded arms, with lanyards wrapped on the side. Enough clearance must be left for the helicopter and the payload hanging on a 44.87m long rope [42].
2. Once the payload is placed on the net, four hydraulic arms extend fully and meet in the middle to wrap the payload in the net. Corner lanyards, gradually released as the arms unfold, are connected in the middle. 2m clearance is left under the payload to be picked up by another vehicle.

To satisfy both cases, hydraulic arms consist of four parts, two of which are 2.33m long, and two of 3.76m. Members of the same length are connected by rotational joints, and two resulting parts are connected by a revolving joint. Each arm ends with a claw holding the lanyard hook. It was estimated that the base of each arm should not be greater than $1.5\text{m} \times 1.5\text{m}$.

Top view of the net and the system together with hydraulic arms is shown in Figure 7. Side and front view schematic drawings are included in Appendix A.4.



(a) Cargo net dimensions, measured in flat.



(b) Complete system layout with hanging cargo net and hydraulic arms included.

Figure 7: Schematic top view drawings of the landing capture mechanism.

5 Conclusion

While the preliminary weight sizing process was successful and the proposed algorithm did converge to a reasonable value when compared to historical trends, its accuracy could be further improved. To obtain more reliable and optimal weight estimates, trade-off studies should be conducted before the start of the iterative sizing process. Based on a detailed analysis of the helicopter weight variation in relation to the blade aspect ratio or hover tip speed, a new, optimum set of fixed inputs could be used to produce a better result.

The unconventional engine location posed a challenge for the inlet and exhaust placement. Final arrangement minimises risk factors and is beneficial for the engine efficiency. Selected oil system guarantees safe engine operation throughout the flight.

A cargo net system was chosen to safely capture the payload in landing. To further evaluate the feasibility of the proposed mechanism, a detailed structural analysis is required to calculate forces acting at each joint and size the hydraulic arms more accurately.

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A Appendices

A.1 Historical Data for Linear Regression

Table 5: *Parameters of existing unmanned helicopters.*

Vehicle name	Payload weight, kg	MTOW, kg	Empty weight, kg	Installed power, kW
MQ-8C Fire Scout	453	2721	1452	522
MQ-8B Fire Scout	272	1430	940	313
A160 Hummingbird	454	1814	1134	410
H-6U Unmanned Little Bird	635	1600	862	317
The Snark	680	1133	480	183
Skeldar V-200	40	230	135	41
CybAero APID 60	20	160	105	41
UVH R22	180	635	399	93

A.2 Inlet and Exhaust Geometry Calculations

For a cruise Mach number M and optimal Mach number at the compressor M_c , the area ratio between the inlet (A_i) and compressor (A_c) can be expressed as

$$\frac{A_i}{A_c} = \frac{A_i}{A^*} \frac{A^*}{A_c}, \quad (4)$$

where A^* is the throat area, and $\frac{A_i}{A^*}$ and $\frac{A^*}{A_c}$ are obtained from 1D Compressible Flow Relations for M and M_c [24].

Turbine exit Mach M_t number is an engine parameter. In order to obtain a desired pressure drop in the exhaust duct, expressed as $\frac{p_e}{p_t}$, where p_t is the turbine exit pressure and p_e is the pressure at the end of the exhaust duct (ambient), a following relationship is used:

$$\frac{p_e}{p_t} = \frac{p_e}{p_0} \frac{p_0}{p_t}, \quad (5)$$

where p_0 is the total pressure. Since $\frac{p_e}{p_t}$ is a set value and $\frac{p_0}{p_t}$ can be obtained from 1D Compressible Flow Relations for M_t , this is rearranged for

$$\frac{p_e}{p_0} = \frac{p_e}{p_t} \frac{p_t}{p_0}, \quad (6)$$

hence exit Mach number M_e is found. The area ratio between the turbine (A_t) and the exhaust (A_e) is then calculated from

$$\frac{A_e}{A_t} = \frac{A_e}{A^*} \frac{A^*}{A_t}, \quad (7)$$

analogically to the method used for the inlet geometry.

A.3 Exhaust Material Selection

Figure 8 shows all materials considered for the exhaust duct after service temperature and corrosion resistance requirements were applied in the CES.

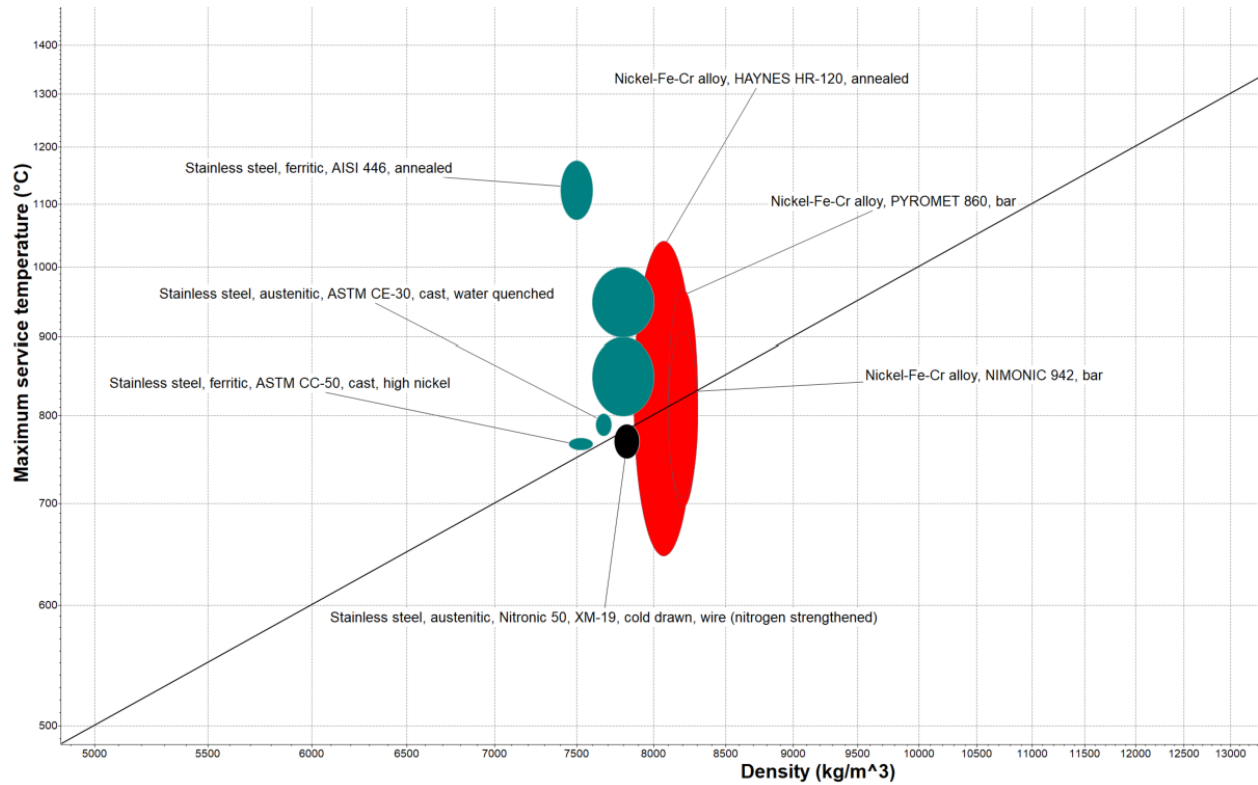


Figure 8: Exhaust duct material selection chart.

A.4 Landing Capture Mechanism Drawings

Figures 9 and 10 shows the front and side view of the proposed landing capture mechanism.

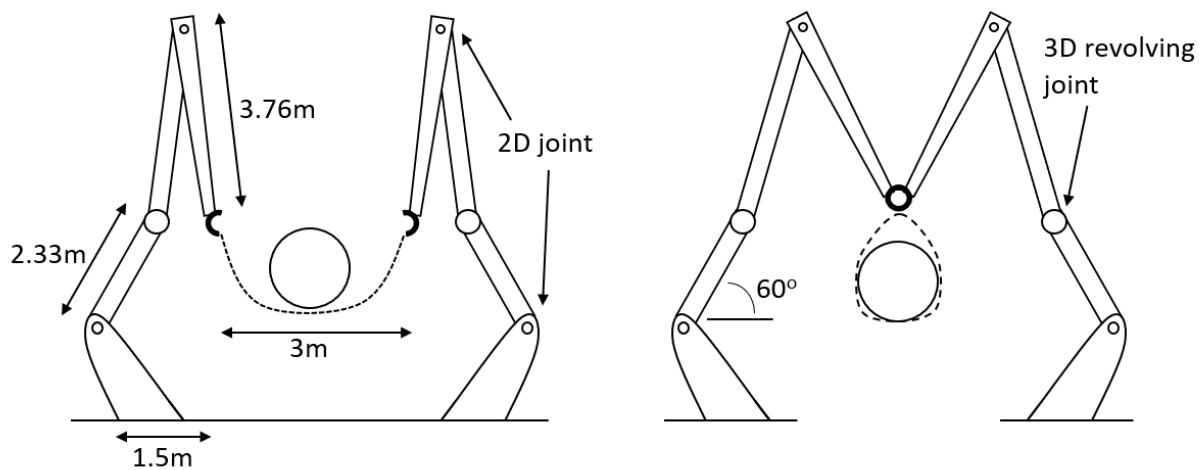


Figure 9: Capture mechanism - front view.

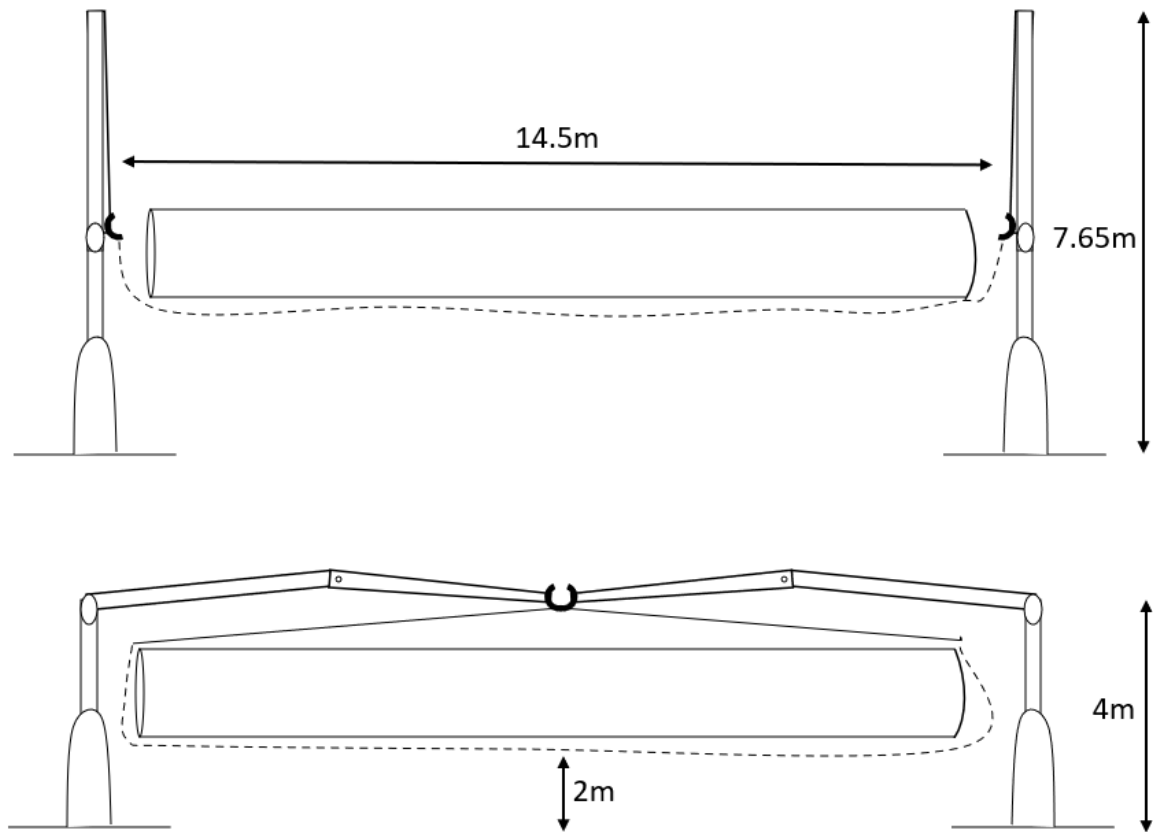


Figure 10: *Capture mechanism - side view.*